Drag Drop Questions 2

10 questions | Original English questions and source answers

Question 1

Drag each route source from the left to the numbers on the right. Beginning with the lowest and ending with the highest administrative distance.

Available items

connected

static

EBGP

EIGRP

OSPF

RIP

Descriptions / target positions

1

2

3

4

5

6

Answer

Description / position	Source answer
1	connected
2	static
3	EBGP
4	EIGRP
5	OSPF
6	RIP

Source explanation

1: connected (AD = 0)
2: static (AD = 1)
3: EBGP (AD = 20)
4: EIGRP (AD = 90)
5: OSPF (AD = 110)
6: RIP (AD = 120)

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 1

Question 2

Order the DHCP message types as they would occur between a DHCP client and a DHCP server.

Available items

DHCPDISCOVER

DHCPOFFER

DHCPREQUEST

DHCPACK

Descriptions / target positions

First

Second

Third

Fourth

Answer

Description / position	Source answer
First	DHCPDISCOVER
Second	DHCPOFFER
Third	DHCPREQUEST
Fourth	DHCPACK

Source explanation

There are four messages sent between the DHCP Client and DHCP Server: DHCPDISCOVER, DHCPOFFER, DHCPREQUEST and DHCPACKNOWLEDGEMENT. This process is often abbreviated as DORA (for Discover, Offer, Request, Acknowledgement).
Answer:
+ First: DHCPDISCOVER
+ Second: DHCPOFFER
+ Third: DHCPREQUEST
+ Fourth: DHCPACK

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 2

Question 3

An interface has been configured with the access list that is shown below.

access-list 107 deny tcp 207.16.12.0 0.0.3.255 any eq http
access-list 107 permit ip any any

On the basis of that access list, drag each information packet on the left to the appropriate category on the right.

Available items

Permitted

Denied

Descriptions / target positions

source IP: 207.16.32.14, destination application: http

source IP: 207.16.15.9, destination port: 23

source IP: 207.16.14.7, destination port: 80

source IP: 207.16.13.14, destination application: http

source IP: 207.16.16.14, destination port: 53

Answer

Description / position	Source answer
source IP: 207.16.32.14, destination application: http	Permitted
source IP: 207.16.15.9, destination port: 23	Permitted
source IP: 207.16.14.7, destination port: 80	Denied
source IP: 207.16.13.14, destination application: http	Denied
source IP: 207.16.16.14, destination port: 53	Permitted

Source explanation

From the ACL:
access-list 107 deny tcp 207.16.12.0 0.0.3.255 any eq http
access-list 107 permit ip any any
We can deduce it block all source hosts in 207.16.12.0/22 range (which ranges from 207.16.12.0 to 207.16.15.255) via HTTP (port 80) while permit any other hosts.

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 3

Question 4

Match the functions to the corresponding layers. (Not all options are used)

Available items

Access layer

Distribution layer

Core layer

Unused

Descriptions / target positions

provides high-speed backbone connectivity

implements personal firewalls on the client computers

provides network access to the user

implements network access policy

represents the network edge

establishes Layer 3 routing boundaries

functions as an aggregator for all the campus blocks

Answer

Description / position	Source answer
provides high-speed backbone connectivity	Core layer
implements personal firewalls on the client computers	Unused
provides network access to the user	Access layer
implements network access policy	Distribution layer
represents the network edge	Access layer
establishes Layer 3 routing boundaries	Distribution layer
functions as an aggregator for all the campus blocks	Core layer

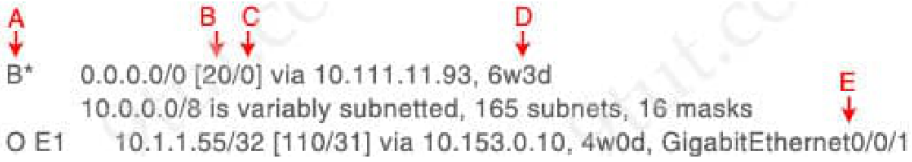
Source explanation

A typical enterprise hierarchical LAN campus network design includes the following three layers:
+ Access layer: Provides workgroup/user access to the network
+ Distribution layer: Provides policy-based connectivity and controls the boundary between the access and core layers
+ Core layer: Provides fast transport between distribution switches within the enterprise campus
Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 4

Question 5

Refer to the exhibit. Drag and drop the routing table components on the left onto the corresponding letter from the exhibit on the right. Not all options are used.



Available items

administrative distance

metric

next-hop interface

outbound interface

route source

subnet mask

timestamp

Descriptions / target positions

A

B

C

D

E

Answer

Description / position	Source answer
A	route source
B	administrative distance
C	metric
D	timestamp
E	outbound interface

Unused items

next-hop interface

subnet mask

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 5

Question 6

Drag the descriptions of device management from the left onto the types of device management on the right.

Available items

Cisco DNA Center Device Management

Traditional Device Management

Descriptions / target positions

monitors the cloud for software updates

manages device configurations on a per-device basis

implements changes via an SSH terminal

uses CLI templates to apply a consistent configuration to multiple devices at an individual location

security is managed near the perimeter of the network with firewalls, VPNs and IPS

uses NetFlow to analyze potential security threats throughout the network and take appropriate action on that traffic

Answer

Description / position	Source answer
monitors the cloud for software updates	Cisco DNA Center Device Management
manages device configurations on a per-device basis	Traditional Device Management
implements changes via an SSH terminal	Traditional Device Management
uses CLI templates to apply a consistent configuration to multiple devices at an individual location	Cisco DNA Center Device Management
security is managed near the perimeter of the network with firewalls, VPNs and IPS	Traditional Device Management
uses NetFlow to analyze potential security threats throughout the network and take appropriate action on that traffic	Cisco DNA Center Device Management

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 6

Question 7

Drag and drop each broadcast IP address on the left to the Broadcast Address column on the right. Not all options are used.

Available items

Broadcast address

Unused

Descriptions / target positions

10.1.255.254/24

10.63.255.255/10

172.16.255.39/29

172.20.255.255/16

192.168.1.10/24

192.168.255.127/25

Answer

Description / position	Source answer
10.1.255.254/24	Unused
10.63.255.255/10	Broadcast address
172.16.255.39/29	Broadcast address
172.20.255.255/16	Broadcast address
192.168.1.10/24	Unused
192.168.255.127/25	Broadcast address

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 7

Question 8

Drag and drop to the characteristics of networking from the left onto the correct networking types on the right.

Available items

Controller-Based Networking

Traditional Networking

Descriptions / target positions

focused on network

focused on devices

user input is a configuration

user input is a policy

uses allow list security model

uses block list security model

Answer

Description / position	Source answer
focused on network	Controller-Based Networking
focused on devices	Traditional Networking
user input is a configuration	Traditional Networking
user input is a policy	Controller-Based Networking
uses allow list security model	Controller-Based Networking
uses block list security model	Traditional Networking

Source explanation

Traditional Networking uses block list model because all traffic is allowed to pass through initially until we configure an ACL, prefix-list… This model is more lenient, because if no application knowledge exists, the default network configuration will let everything through. From a security perspective, it is very dangerous though — every time you allow a protocol that the application does not need, you are unnecessarily increasing your attack surface, and therefore compromising the overall application security.
In Controller-Based Networking, everything is denied by default until we explicitly allow them by a white (allow) list.

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 8

Question 9

Drag and drop the attack-mitigation techniques from the left onto the types of attack that they mitigate on the right.

Available items

configure the native VLAN with a nondefault VLAN ID

configure 802.1x authenticate

configure DHCP snooping

disable DTP

Descriptions / target positions

802.1q double-tagging VLAN-hopping attack

MAC flooding attack

man-in-the-middle spoofing attack

switch-spoofing VLAN-hopping attack

Answer

Description / position	Source answer
802.1q double-tagging VLAN-hopping attack	configure the native VLAN with a nondefault VLAN ID
MAC flooding attack	configure 802.1x authenticate
man-in-the-middle spoofing attack	configure DHCP snooping
switch-spoofing VLAN-hopping attack	disable DTP

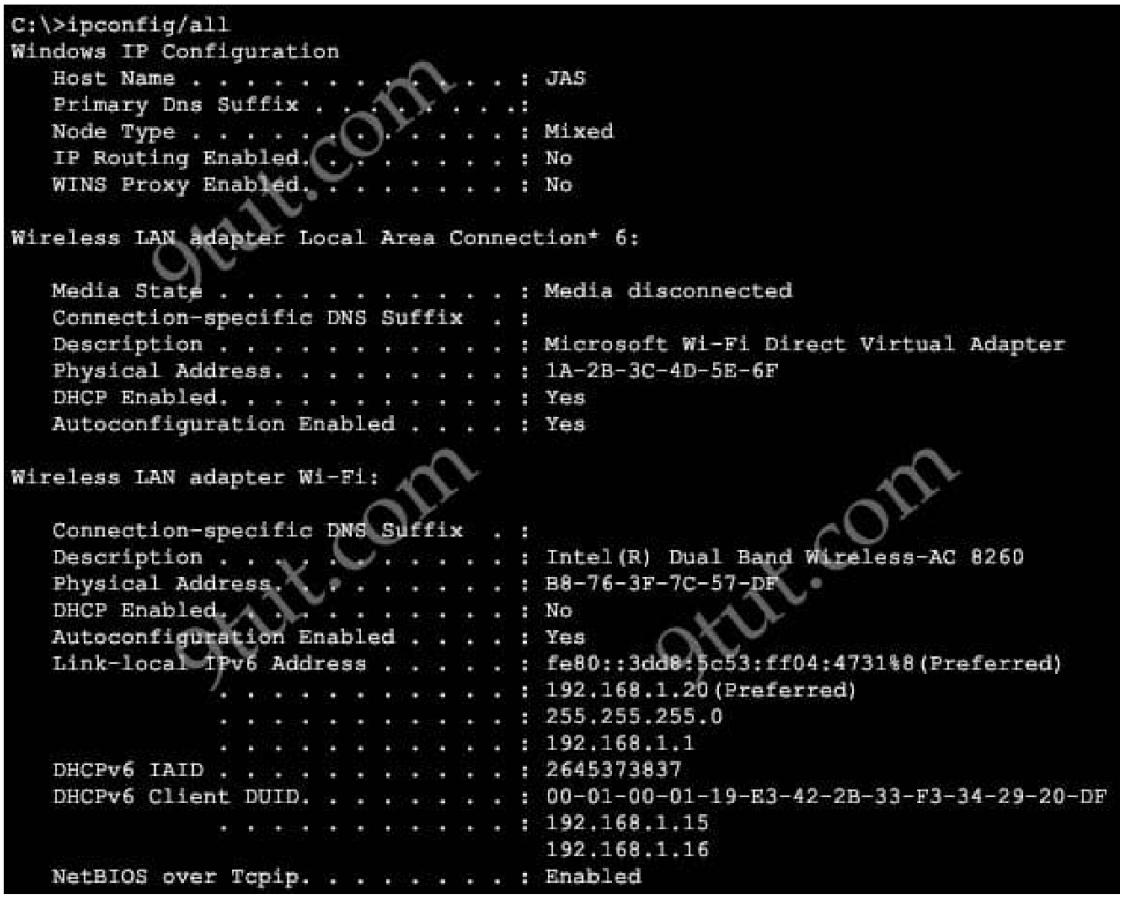
Source explanation

VLAN Hopping: By altering the VLAN ID on packets encapsulated for trunking, an attacking device can send or receive packets on various VLANs, bypassing Layer 3 security measures. VLAN hopping can be accomplished by switch spoofing or double tagging.
a. Switch spoofing:
The attacker can connect an unauthorized Cisco switch to a Company switch port. The unauthorized switch can send DTP frames and form a trunk with the Company Switch. If the attacker can establish a trunk link to the Company switch, it receives traffic to all VLANs through the trunk because all VLANs are allowed on a trunk by default.
(Instead of using a Cisco Switch, the attacker can use a software to create and send DTP frames).
To mitigate this type of attack, we can disable DTP.
b. Double-Tagging attack:
In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).
When the packet from the attacker reaches Switch A, Switch answer " only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch answer " forwards the frame out all links with the same native VLAN 10. Switch answer " receives the frame with an tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.
Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker.
To mitigate this type of attack, we can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers) or implement Private VLANs.
Answer:
+ 802.1q double-tagging VLAN-hopping attack: configure the native VLAN with a nondefault VLAN ID
+ MAC flooding attack: configure 802.1x authenticate
+ man-in-the-middle spoofing attack: configure DHCP snooping
+ switch-spoofing VLAN-hopping attack: disable DTP

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 9

Question 10

An engineer is tasked with verifying network configuration parameters on a client workstation to report back to the team lead. Drag and drop the node identifiers from the left onto the network parameters on the right.



Available items

192.168.1.255

192.168.1.1

192.168.1.20

B8-76-3F-7C-57-DF

192.168.1.254

Descriptions / target positions

broadcast address

default gateway

host IP address

MAC address

last assignable IP address in the subnet

Answer

Description / position	Source answer
broadcast address	192.168.1.255
default gateway	192.168.1.1
host IP address	192.168.1.20
MAC address	B8-76-3F-7C-57-DF
last assignable IP address in the subnet	192.168.1.254

Source explanation

In the output above, three lines under "Link-local IPv6 Address" line have been blacked out so we have to figure out what they are. They are the IP address of the Wireless card and the Default gateway of this computer.
Answer:
+ broadcast address: 192.168.1.255
+ default gateway: 192.168.1.1
+ host IP address: 192.168.1.20
+ MAC address: B8-76-3F-7C-57-DF
+ last assignable IP address in the subnet: 192.168.1.254

Source: CCNA Training » Drag Drop Quiz 2.pdf | Question 10